

Stochastic Reconfiguration in Quantum Monte Carlo

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FYS5419/9419, May 13, 2026

Stochastic reconfiguration

- imaginary-time evolution and variational projection
- tangent-space expansion of the trial state
- derivation of $S \delta \alpha = -\delta \tau \mathbf{g}$
- logarithmic derivatives, metric matrix, force vector
- regularization and algorithmic summary

The score function and Fisher information

- why we use $\log p(x)$
- likelihood, log-likelihood, and the score
- relative sensitivity and information content
- entropy, additivity, and information geometry
- Fisher information matrix; quantum FIM $F_{ij} = 4S_{ij}$
- natural gradient; three-way equivalence with SR
- Cramér–Rao bound and regularization

Setup and notation

We consider a variational wave function

$$|\Psi(\alpha)\rangle, \quad \alpha = (\alpha_1, \alpha_2, \dots, \alpha_p),$$

depending on p real parameters.

The normalized state is

$$|\bar{\Psi}(\alpha)\rangle = \frac{|\Psi(\alpha)\rangle}{\sqrt{\langle\Psi(\alpha)|\Psi(\alpha)\rangle}}.$$

We wish to optimize the parameters such that the energy

$$E(\alpha) = \frac{\langle\Psi(\alpha)|H|\Psi(\alpha)\rangle}{\langle\Psi(\alpha)|\Psi(\alpha)\rangle}$$

is minimized.

Imaginary-time evolution as a route to the ground state

The exact imaginary-time evolution is

$$|\Psi(\tau)\rangle = e^{-\tau H} |\Psi(0)\rangle .$$

Expanding in energy eigenstates $H|n\rangle = E_n|n\rangle$:

$$|\Psi(\tau)\rangle = \sum_n c_n e^{-\tau E_n} |n\rangle .$$

If $c_0 \neq 0$, then for large τ ,

$$|\Psi(\tau)\rangle \propto e^{-\tau E_0} |0\rangle ,$$

so imaginary-time propagation filters out the ground state.

Why a variational projection is needed

In Variational Monte Carlo we restrict to the variational manifold

$$\mathcal{M} = \{|\Psi(\alpha)\rangle \mid \alpha \in \mathbb{R}^p\}.$$

The exact state $(1 - \delta\tau H)|\Psi(\alpha)\rangle$ will in general *not* belong to $\mathcal{M}$.

Idea of stochastic reconfiguration:

- apply one infinitesimal imaginary-time step,
- project the resulting state back onto $\mathcal{M}$,
- interpret the projection as a parameter update $\alpha \rightarrow \alpha + \delta\alpha$.

First-order imaginary-time step

For a small time step $\delta\tau$, $e^{-\delta\tau H} = 1 - \delta\tau H + \mathcal{O}(\delta\tau^2)$.

A convenient norm-preserving first-order approximation is

$$|\Psi'_{\text{exact}}\rangle \approx [1 - \delta\tau(H - E(\alpha))] |\Psi(\alpha)\rangle.$$

Since $\langle\Psi|(H - E)|\Psi\rangle = 0$, the vector $(H - E)|\Psi\rangle$ is orthogonal to $|\Psi\rangle$ to first order, so subtracting $E(\alpha)$ removes the trivial norm-rescaling component and moves the state tangentially in projective Hilbert space.

Tangent-space expansion and projection

The variational state at shifted parameters is

$$|\Psi(\alpha + \delta\alpha)\rangle = |\Psi\rangle + \sum_k \delta\alpha_k |\Psi_k\rangle + \mathcal{O}(\delta\alpha^2), \quad |\Psi_k\rangle \equiv \frac{\partial}{\partial\alpha_k} |\Psi\rangle.$$

The **mismatch vector** $|\Delta\rangle = |\Psi'_{\text{var}}\rangle - |\Psi'_{\text{exact}}\rangle$ has to first order

$$|\Delta\rangle = \sum_k \delta\alpha_k |\Psi_k\rangle + \delta\tau(H - E)|\Psi\rangle.$$

Minimizing $\mathcal{D} = \langle\Delta|\Delta\rangle$ and differentiating with respect to $\delta\alpha_i$:

$$\sum_j \langle\Psi_i|\Psi_j\rangle \delta\alpha_j = -\delta\tau \operatorname{Re} \langle\Psi_i|(H - E)|\Psi\rangle.$$

Logarithmic derivatives and the SR matrix

In configuration space define the **logarithmic derivative**:

$$O_k(x) = \frac{1}{\Psi(x)} \frac{\partial \Psi(x)}{\partial \alpha_k} = \frac{\partial \ln \Psi(x)}{\partial \alpha_k}.$$

Monte Carlo expectation values: $\langle A \rangle = N_s^{-1} \sum_{\ell} A(x^{(\ell)})$, $P(x) = |\Psi(x)|^2 / \langle \Psi | \Psi \rangle$.

Working with normalized states, for real Ψ :

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle \quad (\text{covariance of log-derivatives}),$$

$$g_i = \langle O_i E_{\text{loc}} \rangle - \langle O_i \rangle \langle E_{\text{loc}} \rangle \quad (\text{covariance with local energy}),$$

where $E_{\text{loc}}(x) = (H\Psi)(x)/\Psi(x)$.

Final SR linear system

Collecting everything:

$$S \delta \alpha = -\delta \tau \mathbf{g}, \quad \delta \alpha = -\delta \tau S^{-1} \mathbf{g},$$

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle, \quad g_i = \langle O_i E_{\text{loc}} \rangle - \langle O_i \rangle \langle E_{\text{loc}} \rangle.$$

Comparison with gradient descent: ordinary GD uses $\delta \alpha_{\text{GD}} = -\eta \mathbf{g}$; SR uses $\delta \alpha_{\text{SR}} = -\eta S^{-1} \mathbf{g}$. The inverse metric rescales and mixes parameter directions according to the geometry of the wave-function manifold, giving better conditioning and a direct connection to imaginary-time evolution.

Part II

The Score Function, Fisher Information,
and Their Equivalence with Stochastic Reconfiguration

Why does $\log p(x)$ appear everywhere?

In statistics, machine learning, information theory, and statistical physics, the quantity $\log p(x)$ is ubiquitous.

Where it appears

- Maximum likelihood estimation
- Fisher information
- Cross-entropy loss
- Shannon entropy
- Bayesian inference
- Statistical mechanics
- Variational inference

One-sentence answer

Probabilities *multiply* while optimisation prefers *additive* structures. The logarithm converts products into sums and relative changes into absolute ones.

The following slides make this precise.

Likelihood and why we take its logarithm

Observe independent data $x_1, \dots, x_N$ drawn from $p(x|\theta)$. The **likelihood** is

$$P(D|\theta) = \prod_{i=1}^N p(x_i|\theta).$$

Products are inconvenient: numerically unstable, exponentially small, hard to differentiate.
Taking the logarithm:

$$\log P(D|\theta) = \sum_{i=1}^N \log p(x_i|\theta).$$

Because log is monotonic,

$$\arg \max_{\theta} P(D|\theta) = \arg \max_{\theta} \log P(D|\theta),$$

so maximising the likelihood and the log-likelihood are *equivalent*. The log-likelihood is far easier to work with analytically and numerically.

The score function

The **score function** is the gradient of the log-likelihood:

$$s(\theta) = \partial_{\theta} \log p(x|\theta) = \frac{1}{p(x|\theta)} \partial_{\theta} p(x|\theta).$$

The second form shows that the score measures the **relative sensitivity** of the distribution to the parameter:

$$s(\theta) = \frac{\delta p / \partial \theta}{p}.$$

Why relative, not absolute? Suppose $p(x) \ll 1$. A small absolute change in p may correspond to a large relative change — and it is the relative change that matters for distinguishability. The logarithmic derivative $\delta p / p$ captures the natural geometry of probability distributions.

Key identities of the score function

Identity 1 — zero mean. Differentiating the normalisation $\int p(x|\theta) dx = 1$:

$$\mathbb{E}[\partial_\theta \log p] = \int p \frac{\partial_\theta p}{p} dx = \partial_\theta \int p dx = 0.$$

Identity 2 — Hessian form of Fisher information. Differentiating the zero-mean identity again:

$$\mathbb{E}[(\partial_\theta \log p)^2] = -\mathbb{E}[\partial_\theta^2 \log p].$$

These two identities hold for any smooth parametric family and rely crucially on the logarithm. They are the foundation of both maximum likelihood theory and the Fisher information matrix.

Information content and entropy

Define the **information content** of an event x :

$$I(x) = -\log p(x).$$

Rare events ($p(x) \ll 1$) carry large information; common events carry little.

For independent events A and B , probabilities multiply but information *adds*:

$$p(A, B) = p(A) p(B) \implies I(A, B) = I(A) + I(B).$$

The **Shannon entropy** is the expected information content:

$$S = -\sum_x p(x) \log p(x) = \mathbb{E}[I(x)].$$

Entropy measures the average surprise of the distribution. The logarithm is the *unique* function (up to a constant) that makes information additive for independent events.

Connection to statistical mechanics and machine learning

Statistical mechanics

The Boltzmann distribution is

$$p(x) = \frac{e^{-\beta E(x)}}{Z}.$$

Taking logarithms:

$$\log p(x) = -\beta E(x) - \log Z,$$

so energy is *proportional* to log-probability.
This makes the connection to thermodynamics exact and allows free-energy calculations via log-likelihoods.

Machine learning

Many generative models are built on log-probabilities:

- cross-entropy loss: $L = -\sum_i y_i \log p_i$
- variational autoencoders
- diffusion models
- Boltzmann machines
- Bayesian neural networks

In each case the logarithm converts the product structure of joint probabilities into an additive, differentiable objective.

Fisher information matrix — definition

Let $p(x; \alpha)$ be a parametric family of distributions and $s_k = \partial_{\alpha_k} \log p(x; \alpha)$ the score. Because $\mathbb{E}_p[s_k] = 0$, the **Fisher information matrix** (FIM) equals the covariance of the score:

$$F_{ij} = \mathbb{E}_p[s_i s_j] = \text{Cov}_p(s_i, s_j) = -\mathbb{E}_p \left[\frac{\partial^2 \log p}{\partial \alpha_i \partial \alpha_j} \right].$$

Interpretations:

- **Curvature of the log-likelihood:** large F_{ii} means the likelihood surface is sharply curved around the parameter α_i .
- **Distinguishability:** large F_{ii} means nearby distributions are easy to tell apart.
- **Riemannian metric:** F defines distances between distributions via $ds^2 = d\alpha^\top F d\alpha$.

Fisher information as the curvature of KL divergence

Theorem. For $\alpha' = \alpha + d\alpha$,

$$D_{\text{KL}}(p(\cdot; \alpha) \| p(\cdot; \alpha')) = \frac{1}{2} d\alpha^\top F(\alpha) d\alpha + \mathcal{O}(d\alpha^3).$$

Proof sketch. Taylor-expand the KL divergence to second order in $d\alpha$:

$$D_{\text{KL}}(p \| p') = -\mathbb{E}_p \left[\sum_k d\alpha_k s_k + \frac{1}{2} \sum_{kl} d\alpha_k d\alpha_l \frac{\partial^2 \log p}{\partial \alpha_k \partial \alpha_l} \right] + \mathcal{O}(d\alpha^3).$$

The first-order term vanishes because $\mathbb{E}_p[s_k] = 0$. The second-order term gives $\frac{1}{2} d\alpha^\top F d\alpha$ by the Hessian form. $\square$ Therefore F is the **Fisher–Rao metric** — the Riemannian metric on the statistical manifold induced by the KL divergence.

Geometry of probability space

Probability distributions form a **curved statistical manifold**.

The score function $\partial_\theta \log p(x|\theta)$ defines the tangent vectors to this manifold: they measure how the information content of an observation changes with model parameters.

The **Fisher–Rao metric**

$$ds^2 = \sum_{ij} F_{ij} d\alpha_i d\alpha_j$$

makes this manifold Riemannian. Distances, curvatures, and information measures become mathematically tractable in these coordinates.

Why logarithms?

The logarithm transforms the positivity constraint $p \geq 0$ and normalisation $\int p dx = 1$ into an unconstrained, geometrically flat coordinate system where gradients measure relative changes.

Score for the VMC distribution

In VMC the sampling distribution is $p(x; \alpha) = |\Psi(x; \alpha)|^2 / Z(\alpha)$.

Lemma. Differentiating $Z = \int |\Psi|^2 dx$:

$$\partial_{\alpha_k} \ln Z = 2 \langle O_k \rangle.$$

Therefore the score for the VMC distribution is

$$s_k(x; \alpha) = \partial_{\alpha_k} \log p = 2(O_k(x) - \langle O_k \rangle) \quad (\text{for real } \Psi).$$

The score is the **centred logarithmic derivative** $O_k = \partial_{\alpha_k} \ln \Psi$ — the same object that enters the SR matrix through $S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle$. This is the key bridge between the SR and FIM formalisms.

Quantum Fisher information matrix equals $4S$

Substituting the VMC score into the FIM definition:

$$F_{ij} = \mathbb{E}_p[s_i s_j] = 4 \mathbb{E}_p[(O_i - \langle O_i \rangle)(O_j - \langle O_j \rangle)] = 4(\langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle).$$

$$F_{ij} = 4 S_{ij}.$$

The factor of four arises from the squared coefficient in the score. The SR matrix S is therefore **one quarter of the Fisher information matrix** of the Born distribution $p(x; \alpha)$.

The Fubini–Study metric (natural distance on quantum state space) gives the same result: for real Ψ , $ds_{\text{FS}}^2 = d\alpha^\top S d\alpha$, confirming that S governs both the quantum-state distance and the KL distance between Born distributions.

Natural gradient: derivation via constrained optimisation

The **natural gradient** is the steepest-descent direction in the Fisher–Rao metric: minimize $\mathcal{L}$ subject to a fixed KL step size,

$$\min_{\delta\alpha} \mathbf{g}^\top \delta\alpha \quad \text{s.t.} \quad \delta\alpha^\top F \delta\alpha = \varepsilon^2.$$

Lagrange multiplier μ gives $\mathbf{g} - 2\mu F \delta\alpha = \mathbf{0}$, hence

$$\delta\alpha_{\text{NG}} = -\eta F^{-1} \mathbf{g}.$$

Plain gradient descent is the special case $F = I$ (Euclidean metric on parameter space).

Reparameterisation covariance. Under $\beta = f(\alpha)$ with Jacobian J , the FIM transforms as $F^{(\beta)} = J^\top F^{(\alpha)} J$, so $\delta\beta_{\text{NG}} = J^{-1} \delta\alpha_{\text{NG}}$: the natural-gradient update is *covariant*. Plain gradient descent is not.

Three-way equivalence

Since $F_{ij} = 4S_{ij}$, the natural gradient update $\delta\alpha_{\text{NG}} = -\eta F^{-1}\mathbf{g} = -\frac{\eta}{4}S^{-1}\mathbf{g}$ recovers the SR update when $\eta/4 = \delta\tau$. All three descriptions are the *same* update:

Projected imaginary time

$$S \delta\alpha = -\delta\tau \mathbf{g}$$



Natural gradient (statistics)

$$\delta\alpha = -\frac{\delta\tau}{4} F^{-1}\mathbf{g}$$



Natural gradient (quantum geometry)

$$\delta\alpha = -\delta\tau S^{-1}\mathbf{g}$$

Quantum mechanics (ITE projection), information geometry (steepest descent on the statistical manifold), and optimisation theory (natural gradient) all say the same thing.

Cramér–Rao bound and principled regularization

The FIM controls **how precisely parameters can be estimated**.

Cramér–Rao bound: for any unbiased estimator $\hat{\alpha}$ from N_s i.i.d. samples,

$$\text{Cov}(\hat{\alpha}) \succeq \frac{1}{N_s} F(\alpha)^{-1}.$$

In VMC: directions with large F_{kk} can be estimated precisely and updated aggressively; directions with small F_{kk} cannot be estimated beyond the CR limit and should be damped.

This gives a **principled interpretation of regularization**:

$$(S + \lambda I)^{-1} = \left(\frac{1}{4}F + \lambda I\right)^{-1}$$

damps parameter directions below the Cramér–Rao estimation floor, interpolating between SR ($\lambda \rightarrow 0$) and plain gradient descent ($\lambda \rightarrow \infty$).

Geometry and connections summary

Object	Expression	Geometric / statistical role
Score function	$s_k = \partial_{\alpha_k} \log p$	relative sensitivity; tangent to stat. manifold
Information content	$I(x) = -\log p(x)$	additive; large for rare events
Shannon entropy	$S = -\mathbb{E}[\log p]$	expected information content
Fisher information matrix	$F_{ij} = \mathbb{E}[s_i s_j]$	curvature of log-likelihood
Fisher–Rao metric	$ds^2 = d\alpha^\top F d\alpha$	distance on statistical manifold
KL divergence (2nd order)	$D_{\text{KL}} \approx \frac{1}{2} d\alpha^\top F d\alpha$	KL curvature $\equiv$ Fisher metric
VMC score	$s_k = 2(\hat{O}_k - \langle O_k \rangle)$	centred log-derivative
SR matrix vs FIM	$F_{ij} = 4S_{ij}$	quantum Fisher = $4 \times$ SR metric
Fubini–Study metric	$ds_{\text{FS}}^2 = d\alpha^\top S d\alpha$	quantum state distance
Natural gradient	$\delta\alpha = -\eta F^{-1} \mathbf{g}$	steepest descent on stat. manifold
SR update	$S \delta\alpha = -\delta\tau \mathbf{g}$	projected ITE $\equiv$ natural gradient
Cramér–Rao bound	$\text{Cov}(\hat{\alpha}) \succeq (N_s F)^{-1}$	estimation precision limit

Regularization and practical solution

In practice S may be singular or ill-conditioned because it is estimated from a finite Monte Carlo sample.

A standard fix is diagonal regularization:

$$S \rightarrow S + \lambda I, \quad \lambda > 0.$$

Then solve

$$(S + \lambda I) \delta \alpha = -\delta \tau \mathbf{g}.$$

Interpretation:

- stabilizes inversion,
- damps directions with negligible Fisher information (flat directions of the statistical manifold),
- interpolates between SR ($\lambda \rightarrow 0$) and ordinary gradient descent ($\lambda \rightarrow \infty$).

Algorithmic summary

At each iteration:

- 1 Sample configurations $x^{(\ell)}$ from $P(x) = |\Psi(x)|^2 / \langle \Psi | \Psi \rangle$.
- 2 Compute logarithmic derivatives $O_k(x^{(\ell)}) = \partial_{\alpha_k} \ln \Psi(x^{(\ell)})$.
- 3 Compute local energies $E_{\text{loc}}(x^{(\ell)}) = (H\Psi)(x^{(\ell)}) / \Psi(x^{(\ell)})$.
- 4 Estimate

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle, \quad g_i = \langle O_i E_{\text{loc}} \rangle - \langle O_i \rangle \langle E_{\text{loc}} \rangle.$$

- 5 Solve $(S + \lambda I) \delta \alpha = -\delta \tau \mathbf{g}$.
- 6 Update $\alpha \leftarrow \alpha + \delta \alpha$.

Stochastic reconfiguration

- ITE selects the ground state; projecting onto $\mathcal{M}$ gives $S \delta \alpha = -\delta \tau \mathbf{g}$.
- S_{ij} : covariance of log-derivatives.
- g_i : covariance with the local energy.

Why $\log p(x)$ is fundamental

- Converts products into sums.
- Measures relative sensitivity $\delta p/p$.
- Makes information additive.
- Connects energy to probability (stat. mech.).
- Defines the natural geometry on probability space.

Fisher information and natural gradient

- Score $s_k = \partial_{\alpha_k} \log p$ has zero mean; its covariance is the FIM.
- FIM = curvature of KL divergence = Fisher–Rao metric.
- For VMC: $F_{ij} = 4S_{ij}$; SR matrix S is $\frac{1}{4}$ of the FIM.
- SR update = natural gradient = projected imaginary-time evolution.
- Natural gradient is reparameterisation-covariant; GD is not.
- Regularization $(S + \lambda I)^{-1}$ damps directions below the Cramér–Rao floor.